Related Work Table

No.	Title	Authors	Year	Dataset	Application area	Method / model	Metrics	Strengths	Limitations	Research gap	Relation to our work
1	MedNet: A Lightweight Attention-Augmented CNN for Medical Image Classification	Md. Ferdous, Saifuddin Mahmud, Md. Eleush Zahan Shimul	2025	DermaMNIST, BloodMNIST, OCTMNIST which from MedMNIST, and Fitzpatrick 17k datasets	Medical-image classification and blood-cell recognition	MedNet: Lightweight CNN using depthwise-separable convolution, residual connections and CBAM channel-spatial attention.	Accuracy, AUC, precision, recall, F1-score and computational cost. 0.998 ± 0.0015 AUC and 0.983 ± 0.0047 accuracy on BloodMNIST	Uses the MedNet which include BloodMNIST dataset, strong classification performance; lightweight architecture; attention emphasizes important spatial and channel features.	Does not model relationships between different image regions using graph convolution.	CNN-attention has been evaluated on BloodMNIST, but attention-enhanced features have not been converted into graphs for GCN learning.	Both use BloodMNIST and attention-based CNN classification. Our work extends it by adding GCN-based graph learning.

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2	White Blood Cell Classification Using Multi-Hop Attention Graph Neural Networks	Minh Ly Duc, Petr Bilik, Radek Martinek	2025	16,027 annotated high-resolution WBC images from 9 classes	White-blood-cell classification and leukemia diagnosis support	YOLO-v10 and CentreNet for image processing, SSO for feature selection and MAGNA Multi-Hop Attention GNN for classification. Image regions are graph nodes, while distance or similarity relationships form edges.	Accuracy, sensitivity, specificity, PPV, NPV and F1-score. YLSSOGNN achieved 99.18% accuracy, while YLGNN achieved 99.03%.	Uses graph attention to capture both local and distant relationships, and achieves strong WBC classification performance.	Uses a complex YOLO–CentreNet–SSO–GNN pipeline, not tested on BloodMNIST. computationally heavy for 28×28 images.	A simpler image-to-graph attention-GCN framework for low-resolution	Both studies classify blood-cell images using graph-based learning, their work uses WBC data, while ours applies CNN–Attention–GCN to BloodMNIST.

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3	Fuzzy Attention Convolutional Neural Networks: A Novel Approach Combining Intuitionistic Fuzzy Sets and Deep Learning	Zheng Zhao, Doo Heon Song, Kwang Baek Kim	2026	PathMNIST-30000, full PathMNIST, and BloodMNIST	Medical-image classification and uncertainty-aware feature learning	CNN with Fuzzy Attention Network Layer, using membership (μ), non-membership (ν), and hesitation (π) values and cross-layer attention guidance. No GCN or graph construction was used.	Accuracy, recall, F1-score, robustness, calibration and ablation. Reported 84.41 ± 0.56% accuracy, 1.69% improvement over baseline and ECE = 0.0452; removing components reduced performance by about 1.7–2.0%.	Explicitly models uncertainty in feature importance, lightweight, supports interpretable and calibrated predictions.	Does not model spatial relationships using GCN. fuzzy-clustering and cross-layer mechanisms increase design complexity.	Fuzzy attention has been evaluated on BloodMNIST, but it has not been integrated with explicit image-graph construction and GCN learning.	Relevant because it uses BloodMNIST and attention-based feature learning; our work combines attention with CNN and GCN.

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4	Hybrid Convolution Neural Network and Graph Neural Network for Detection of Leukocyte	Khalid Altarawneh	2023	Approximately 12,500 augmented microscopic blood-cell images with four cell types-eosinophil, lymphocyte, monocyte and neutrophil classes	Leukocyte detection and classification	Hybrid CNN + GNN model. CNN extracts local image features, while the GNN is intended to learn contextual relationships.	Accuracy, precision, recall, F1-score and confusion matrix. The proposed model achieved 96.87% accuracy.	Combines local CNN features with graph-based contextual learning; directly related to blood-cell image classification.	Uses only four classes; lacks an attention module; graph nodes, edges, adjacency construction and message-passing procedure are insufficiently documented. The specific graph construction and GNN layer type are not clearly explained.	A reproducible CNN-GCN model with explicit graph construction and attention is still required for multiclass blood-cell classification.	Relevant because both combine CNN and graph learning for blood-cell classification; our work additionally uses attention and BloodMNIST.

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5	Implementing Vision Transformer for Classifying 2D Biomedical Images	Arindam Halder, Sanghita Gharami, Priyangshu Sadhu, Pawan Kumar Singh, Marcin Woźniak, Muhammad Fazal Ijaz	2024	MedMNISTv2 dataset: BloodMNIST, BreastMNIST, PathMNIST and RetinaMNIST	Multiclass biomedical-image classification	Pretrained ViT-Base-Patch16-224 using image patches and multi-head self-attention	Accuracy, precision, recall, F1-score and confusion matrix. BloodMNIST accuracy was 97.90%.	Uses the exact BloodMNIST dataset; self-attention captures long-range relationships between image patches; strong benchmark result.	Does not use CNN or GCN; resizing 28×28 images to 224×224 increases computation without adding original image information.	Self-attention has been applied to BloodMNIST, but CNN-based local features and explicit graph relationships have not been jointly modelled.	Relevant because it uses BloodMNIST with attention-based learning; our work applies CNN, attention and GCN to the same dataset. .

References

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